

Suggested solutions

1	(a)	(i)	From the measured width of the cubes, take average. Compare the <u>average value to the expected value of 2.0 cm</u>; the <u>closer it is to 2.0 cm, the more accurate</u> is the dimension of the cubes.	M1 A1
1	(a)	(ii)	From the measured width of the cubes, <u>compare the widths of each cube to one another</u>. The <u>closer the widths of each cube, the more precise is the dimensions</u> of his printed cubes.	M1 A1
1	(a)	(iii)	He can take <u>repeated measurements of each edge at different positions</u> and take the <u>average of the measurements</u>. OR He can take <u>measurements of the widths in the x,y,z directions</u> and take the <u>average of the measurements</u>.	B1 B1 B1 B1
1	(b)		mass = (length × width × height) × density = (1.99 × 1.95 × 2.02) × 1.24 [M1] = 9.7199 = 9.71 g By calculating percentage / fractional uncertainty: $\frac{\Delta m}{m} = \frac{\Delta \ell}{\ell} + \frac{\Delta w}{w} + \frac{\Delta h}{h} + \frac{\Delta \rho}{\rho}$ $= 0.01 + 0.01 + 0.01 + \frac{0.05}{1.24} \quad [*M1]$ = 0.070323 $\Delta m = 0.070323 \times 9.7199 \quad [*M1]$ = 0.68353 = 0.7 g mass = (9.7 ± 0.7) g [A1] Alternatively, max-mean method: $m_{\max} = (1.99 \times 1.01)(1.95 \times 1.01)(2.02 \times 1.01)(1.29) \quad [*M1]$ = 10.418 g $\Delta m = m_{\max} - m_{\text{mean}} = 10.418 - 9.7199 = 0.7 \text{ g} \quad [*M1]$	